

Equality and endpoint Functional Ghobber–Jaming principles

Candidate full solution to Questions 2.10–2.11 of arXiv:2306.01014

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Result. For $1 < p < \infty$, the source inequality has equality only at the zero vector. For $p = 1$ and $p = \infty$, all changes of p -orthonormal basis are generalized permutations; this gives a necessary-and-sufficient support condition, sharp endpoint estimates, and their equality cases.

1 Source questions and notation

Let $(f_j, \tau_j)_{j=1}^n$ and $(g_k, \omega_k)_{k=1}^n$ be p -orthonormal bases of an n -dimensional Banach space X over $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$. Thus the coordinate maps

$$J_f x = (f_j(x))_{j=1}^n, \quad J_g x = (g_k(x))_{k=1}^n$$

are onto linear isometries from X to ℓ_p^n . Put

$$U = J_g J_f^{-1} : \ell_p^n \longrightarrow \ell_p^n, \quad U_{kj} = g_k(\tau_j).$$

For $S \subset [n]$, let P_S be the coordinate projection onto S .

For $1 < p < \infty$, let q be conjugate to p , and define

$$c = \max_{j,k} |g_k(\tau_j)|, \quad \alpha = |M|^{1/q} |N|^{1/p} c.$$

The source's Theorem 2.5 assumes $\alpha < 1$ and proves

$$\|x\| \leq C_\alpha (A(x) + B(x)), \quad C_t = 1 + \frac{1}{1-t}, \tag{1}$$

where

$$A(x) = \|P_{M^c} J_f x\|_p, \quad B(x) = \|P_{N^c} J_g x\|_p.$$

Question 2.10 on source PDF page 6 asks for the equality cases in (1). Question 2.11 on PDF page 7 asks for versions at $p = 1$ and $p = \infty$ [1].

2 Idea of the proof

There are two short rigidity mechanisms.

First, retain the exact restricted transition norm

$$s = \|P_N U P_M\| < 1$$

instead of immediately replacing it by the coarser bound $s \leq \alpha$. The source argument yields an asymmetric estimate whose coefficient of the M -tail is strictly smaller than C_s . Hence a nonzero

equality vector would have to have zero N -tail. In ℓ_p , the two coordinate pieces of a vector satisfy a p -power identity. This makes the reverse-triangle step strict by

$$(1 - s^p)^{1/p} > 1 - s \quad (0 < s < 1),$$

and eliminates the last possible equality case.

Second, at either endpoint, a surjective isometry of ℓ_p^n is a generalized permutation. The uncertainty problem therefore becomes purely combinatorial: after transporting one proposed support set by the transition permutation, the two supports must be disjoint. If they overlap, a coordinate vector defeats every tail estimate; if they are disjoint, the complements cover all coordinates and give the sharp inequalities immediately.

3 No nonzero equality cases for $1 < p < \infty$

Theorem 1 (Complete answer to Question 2.10). *Assume $1 < p < \infty$ and $\alpha < 1$. Equality holds in the source inequality (1) if and only if $x = 0$.*

Proof. Write $z = J_fx$, $E = P_M$, $F = P_N$, and

$$s = \|FUE\| \leq |M|^{1/q}|N|^{1/p}c = \alpha < 1.$$

Set $G = \|(I - F)Uz\|_p$ and $H = \|(I - E)z\|_p$; these are respectively the g -tail and the f -tail (so $G = B(x)$ and $H = A(x)$ in the notation of (1)). For the M -part,

$$(1 - s)\|Ez\|_p \leq \|(I - F)UEz\|_p \quad (2)$$

$$\leq \|(I - F)Uz\|_p + \|(I - F)U(I - E)z\|_p \leq G + H. \quad (3)$$

Consequently

$$\|z\|_p \leq \frac{G}{1 - s} + \left(1 + \frac{1}{1 - s}\right)H \leq C_s(G + H) \leq C_\alpha(G + H). \quad (4)$$

Let $z \neq 0$. If $G > 0$, the middle inequality in (4) is strict, because its coefficient of G was increased by exactly one. Suppose $G = 0$. Then $H > 0$: otherwise $z = Ez$ and $Uz = FUEz$, forcing $\|FUE\| \geq 1$.

If $s < \alpha$, the last inequality in (4) is strict. It remains to consider $s = \alpha$. When $\alpha = 0$, one of M, N is empty. If M is empty then $H = \|z\|_p$ and $C_0 = 2$; if N is empty then $G = \|Uz\|_p \neq 0$. Either way equality is impossible.

Thus suppose $s = \alpha \in (0, 1)$. If $Ez = 0$, then again $H = \|z\|_p$ and the estimate is strict. Otherwise the disjoint coordinate pieces of UEz give

$$\begin{aligned} \|(I - F)UEz\|_p^p &= \|Ez\|_p^p - \|FUEz\|_p^p \\ &\geq (1 - s^p)\|Ez\|_p^p. \end{aligned}$$

Since $G = 0$,

$$(I - F)UEz = -(I - F)U(I - E)z,$$

so

$$H \geq \|(I - F)UEz\|_p \geq (1 - s^p)^{1/p}\|Ez\|_p > (1 - s)\|Ez\|_p.$$

Therefore

$$\|z\|_p \leq \|Ez\|_p + H < \frac{H}{1 - s} + H = C_s H = C_\alpha(G + H).$$

Every nonzero vector gives strict inequality; $x = 0$ plainly gives equality. $\square$

4 The endpoint classification

Extend the definition of a p -orthonormal basis verbatim to $p = 1$ and $p = \infty$, replacing the coefficient norm by ℓ_1 or ℓ_∞ .

Lemma 1 (Endpoint isometry rigidity). *Every surjective linear isometry of ℓ_1^n or ℓ_∞^n is a generalized permutation: there are a permutation σ and unimodular scalars ε_j such that*

$$Ue_j = \varepsilon_j e_{\sigma(j)}.$$

Consequently, for two endpoint-orthonormal bases,

$$g_k(\tau_j) = \varepsilon_j \mathbf{1}_{\{k=\sigma(j)\}}.$$

Proof. For ℓ_1^n , put $u_j = Ue_j$. For $j \neq k$, isometry gives $\|u_j + \lambda u_k\|_1 = 2$ for every unimodular λ . Equality in the coordinatewise triangle inequalities for all λ forces u_j and u_k to have disjoint supports. The n nonzero columns therefore have pairwise disjoint supports inside an n -point set, so every support is a singleton and the nonzero entry is unimodular. For ℓ_∞^n , apply the ℓ_1 result to the adjoint isometry on the dual. $\square$

Let $D = \sigma^{-1}(N)$. Thus D is the second proposed support written in the first coordinate system, and

$$\sigma(M) \cap N = \emptyset \iff M \cap D = \emptyset.$$

Theorem 2 (Complete answer to Question 2.11). *Let $p \in \{1, \infty\}$ and let two p -orthonormal bases be related by σ as in Lemma 1. A finite constant C can satisfy a Ghobber–Jamming tail estimate*

$$\|x\| \leq C(A_p(x) + B_p(x)) \quad (x \in X) \quad (5)$$

if and only if $\sigma(M) \cap N = \emptyset$, where

$$A_1 = \sum_{j \in M^c} |f_j(x)|, \quad B_1 = \sum_{k \in N^c} |g_k(x)|,$$

and

$$A_\infty = \max_{j \in M^c} |f_j(x)|, \quad B_\infty = \max_{k \in N^c} |g_k(x)|,$$

with a maximum over the empty set equal to zero.

When the condition holds, the sharp-form estimates are

$$\|x\|_1 \leq A_1(x) + B_1(x), \quad (6)$$

$$\|x\|_\infty = \max\{A_\infty(x), B_\infty(x)\}. \quad (7)$$

If $M \cup D \neq \emptyset$, the best constant in the plus-form (5) is one for both endpoints. If $M = D = \emptyset$, the best plus-form constant is $1/2$.

Proof. Write $z = J_fx$. Up to unimodular factors, $J_gx = Uz$ is the permutation of z by σ . If $M \cap D \neq \emptyset$, choose j in the intersection and take $z = e_j$. Then $A_p = B_p = 0$ while $\|z\|_p = 1$, so no finite C exists.

Suppose $M \cap D = \emptyset$. Every coordinate index lies in M^c or in D^c . For $p = 1$, the sum $A_1 + B_1$ counts every $|z_j|$ at least once, which proves (6). For $p = \infty$, a coordinate attaining $\|z\|_\infty$ belongs to at least one of those complements, proving the identity (7).

If $M \cup D$ is nonempty, a coordinate vector indexed by that union is counted in exactly one tail, proving that plus-form constant one is optimal. If both sets are empty, both tails equal $\|x\|_p$, so the optimal constant is $1/2$. $\square$

Corollary 1 (Endpoint equality cases). *Assume $M \cap D = \emptyset$ and $M \cup D \neq \emptyset$. Equality in (6) holds exactly when*

$$\text{supp}(J_f x) \subseteq M \cup D.$$

Equality in the constant-one plus form at $p = \infty$ holds exactly when $A_\infty(x) = 0$ or $B_\infty(x) = 0$, equivalently when

$$\text{supp}(J_f x) \subseteq M \quad \text{or} \quad \text{supp}(J_f x) \subseteq D.$$

The max-form identity (7) is equality for every x .

Proof. In the ℓ_1 sum, indices outside $M \cup D$ are counted twice and all others once. At $p = \infty$, (7) shows that $A_\infty + B_\infty = \|x\|_\infty$ exactly when one tail vanishes. $\square$

5 Consequences, checks, and novelty scope

At either endpoint the transition coherence is necessarily $\max_{j,k} |g_k(\tau_j)| = 1$. Hence a literal endpoint reading of the source cardinality condition becomes $|N| < 1$ for $p = 1$ and $|M| < 1$ for $p = \infty$. The exact disjointness criterion in Theorem 2 is both sharper and the only possible nontrivial replacement.

The argument is finite-dimensional and applies over both real and complex scalars. It concerns exactly the source's basis notion; it does not assert an endpoint theorem for general frames, where transition operators need not be generalized permutations.

The included verifier tested 209,900 admissible Hilbert-space instances of strict inequality and checked 873,760 endpoint support configurations over random generalized permutations up to dimension seven. These checks support, but are not used in, the proof.

A bounded search through 17 August 2026 used the exact paper title, both question texts, endpoint-basis terminology, and equality keywords. It found the source and unrelated equality results for other uncertainty principles, but no later answer to Questions 2.10–2.11. The only external mathematical input beyond the source is the elementary finite-dimensional endpoint isometry classification, proved above.

References

- [1] K. Mahesh Krishna, *Functional Ghobber–Jamming Uncertainty Principle*, arXiv:2306.01014v1 (2023).